

Effect of osteopathic manipulative treatment on pain, flexibility, autonomic modulation of heart rate, energy and thermal profile in patients with chronic low back pain. blind randomized clinical trial

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ABSTRACT

Introduction: Chronic low back pain (CLBP) is the most common symptom experienced in all ages, being related to incapacity, reduction of life and work quality, having a complex and challenging treatment approach. Osteopathic manipulative treatment (OMT) proposes a global manual intervention in chronic pain.

Objective: To investigate OMT effects on pain, flexibility, autonomic nervous system, energetic and thermal profile in chronic low back pain.

Material and methods: Pain was evaluated by the visual analogue scale (VAS) and algometry, flexibility by the Wells bench, thermal profile of the lumbar spine by infrared thermography, autonomic modulation by heart rate variability (HRV), and the energetic profile by bioelectrography. Osteopathic manipulative treatment group (OMTG) was submitted to three sessions based on osteopathic evaluation and reevaluated immediately after the first session and one week after the last session. The placebo group (PG) got a simulated TENS for the same period.

Results: A significant interaction was observed (ANOVA) between group x time for the flexibility ($F = 7.5$; $p = 0.001$) and VAS ($F = 2.8$; $p = 0.05$), low frequency ($F = 3.5$; $p = 0.04$), low frequency/high frequency ($F = 3.6$; $p = 0.04$) and total energy ($F = 4.4$; $p = 0.01$). A significant increase of flexibility, EVA, L3 algometry and total energy was observed after the intervention for OMTG. Significant reduction of low frequency (LF) and low frequency/high frequency (LF/HF) post-intervention for OMTG.

Conclusion: OMT showed positive effects for pain reducing, increasing flexibility, parasympathetic activities and total energy in patients with CLBP.

1. Introduction

The most common musculoskeletal complaint is chronic low back pain, which has a multifactorial origin, commonly called nonspecific chronic low back pain, is located between the end of the rib cage and the buttock region and can radiate to the lower limbs and can be experienced in all age groups. Generating a high number of disabilities and consequent reduction in quality of life, social participation, work, financial prosperity, involving high health costs, thus considering a globalized health concern (Hartvigsen et al., 2018; De David et al.,

2020).

Its higher prevalence is associated with variables such as female sex, high levels of anxiety and depression, poor sleep quality, poor self-perception of health, smoking, obesity, insufficient levels of physical activity, people exposed to repetitive movements and dissatisfied with work (Falavigna et al., 2015; Gonzales et al. 2021).

Highlighting its multifactorial origin, several variables are related, such as physical factors, a significant relationship was observed between pain and reduced strength of the lumbar extensor and flexor muscles, as well as asymmetry in the range of motion of the hip joint (Shin, 2020)

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and the reduction of shear stress of the thoracolumbar fascia (Langevin et al., 2011; Ribeiro et al., 2018; Kondrup et al., 2022). The individual's psychological and social factors such as kinesiophobia, belief in pain, belief in fear avoidance, self-efficacy, anxiety, coping styles, sleep quality, control and catastrophizing, occupation and demographic characteristics (Agnus Tom et al., 2022), neurological factors, the relationship between chronic pain and central and peripheral sensitization and the lower vagal activity observed by heart rate variability (HRV) (Koenig et al., 2016).

In this scenario, it is understood that the treatment of patients with chronic low back pain remains a major challenge for the health sector (Gonzales et al. 2021) and that it must be carried out taking into account several variables. A wide-ranging option is osteopathic treatment, which has as one of its objectives, balancing the patient in their different models (biomechanical, respiratory-circulatory, metabolic-energetic, neurological, and behavioral). The interaction between these models allows the implementation of an approach that involves not only the physical and physiological aspects, but also the psychological, social, cultural and behavioral aspects in the osteopath/patient relationship, becoming a relevant approach in the management of chronic pain (Seffinger et al., 2018).

The clinically relevant effects of OMT to reduce pain and improve functional status in patients with acute and chronic nonspecific low back pain have already been demonstrated (Franke et al., 2014), as well as its effects on increasing parasympathetic activity (Cerritelli et al., 2021), improved mobility (Zago et al., 2020), improved immune system activity (Licciardone et al., 2012), however, more research is needed to understand the mechanisms for the effects of OMT (Price, 2021), as well as its action in the short and long term (Task Force on the Low Back Pain Clinical Practice Guidelines 2016).

In addition, previous studies have shown changes in the energy profile assessed using Biofield by electrophotonic imaging after OMT (Korotkov et al., 2012). The evaluation and quantification of the Biofield by electrophotonic imaging seeks to investigate the emotional and general health status of the individual, and its variations of increased fields point to beneficial changes in autonomic modulation (Korotkov et al., 2012), however the sample was composed of healthy individuals and its variations in patients with chronic pain need to be investigated. In addition, changes in skin temperature were observed in patients who received manual treatment in the sacral region (WU et al., 2009), as well as after OMT in chronic low back pain (Polidori et al., 2018). These studies point to a reduction in temperature associated with a reduction in tissue stress and the local inflammatory process, however the sample was limited and more studies need to investigate the effects of OMT on thermal modification in chronic pain.

Based on the above, our hypothesis is that OMT can contribute to reducing pain, increasing flexibility, thus interfering with the increase in parasympathetic activity in the autonomic modulation of heart rate, increasing biofield energy by electrophotonic imaging and changing the temperature of the lumbar spine in this condition. The objective of this study was to investigate the effect of OMT through three intervention sessions, on the musculoskeletal system (pain, flexibility and thermal profile), the autonomic nervous system (autonomic heart rate modulation) and the energy profile in patients with chronic low back pain.

2. Methodology

2.1. Study design and ethical aspects

This was a interventional, prospective, randomized and blind study, which followed the recommendations for experimental research with human beings (CNS Resolution 466/12), being approved by the Research Ethics Committee of the Methodist University of Piracicaba – (UNIMEP) by technical advice 3.43.600 (Appendix 1) and registered in the Brazilian Clinical Trials Registry (REBEC) under code RBR-3c8wwt. Volunteers who were studied agreed to participate in the study and

signed the free and informed consent form.

2.2. Subjects

Thirty volunteers with chronic low back pain were screened, who met the following inclusion criteria which were both sexes, disability resulting from low back pain assessed by Oswestry, chronic pain for more than three months, age between eighteen and fifty years old, and availability to participate in the sessions of treatment. The exclusion criteria adopted were the use of medications that interfered with the experimental protocol, respiratory and/or cardiovascular diseases, smoking, presence of tumors of any nature, being under drug treatment for low back pain or other symptoms and depression, having been subjected to any surgical procedure on the lumbar spine, patients with systemic, neurological or rheumatological diseases, pregnancy, undergoing physiotherapeutic treatment for low back pain, or having undergone osteopathic treatment in the last three months. Fig. 1 presents the study volunteer participation flowchart.

2.3. Experimental design

The volunteers were randomized into two groups, osteopathic manipulative treatment (OMTG) and placebo group (PG), subjected to three assessments. The first in the pre-experimental protocol period, the second immediately after the first session and the last assessment one week after the last treatment session. Fig. 2 presents the experimental design of the study. The entire evaluation protocol was carried out by a trained researcher, with experience in collecting the measurements, and blinded as to the group to which the volunteers belonged.

2.4. Assessment of patient characteristics

Initially, an anamnesis was carried out, in which the volunteers were questioned regarding their personal characteristics (age, weight, height), use of medication, treatment, and duration of pain. Information from the anamnesis was used to identify volunteers and apply the study criteria. After the anamnesis, sample characterization questionnaires were carried out by the Oswestry Disability Index (Vigatto et al., 2007), Fear Avoidance Beliefs Questionnaire (Abreu et al., 2008), Start Back Screening Toll (Pilz et al., 2017) and the International Physical Activity Questionnaire (IPAQ).

2.5. Measurement variables

Analogic visual scale was used to measure pain. Consists of a 10-cm line, where the left end corresponds to level 0 (zero) and the right end corresponds to level 10 (ten). To use it, we ask the volunteer to mark on the scale which position corresponds to their pain at that moment, with level 0 corresponding to no pain and level 10 corresponding to the worst pain imaginable. To obtain the result, we must measure with a ruler how many centimeters the pain mark is from level zero (Shafshak and Elneimr, 2021).

Algometry was used to measure the pressure pain test, using perpendicular and gradual pressure exerted on the regions of the spinous processes of the lumbar vertebrae, across an Instrutherm portable digital dynamometer, model DD-200. Volunteers were instructed to inform to the evaluator when the sensation of compression changes to a painful sensation. At the moment that volunteer reported the beginning of the painful sensation, the compression was removed and the value shown on the dynamometer display was noted. To perform the collections, volunteers remained in the prone position, and each point was compressed twice, with a 3-min interval between compressions. As a result, the average value (Kgf) of the two compressions was used. (Liebano et al., 2013; Farasyn and Lassat, 2016; Barrassi et al. 2022).

Posterior chain flexibility was carried out by testing with Wells Bench. Wells Bench consists of a rectangular wooden bench, with an

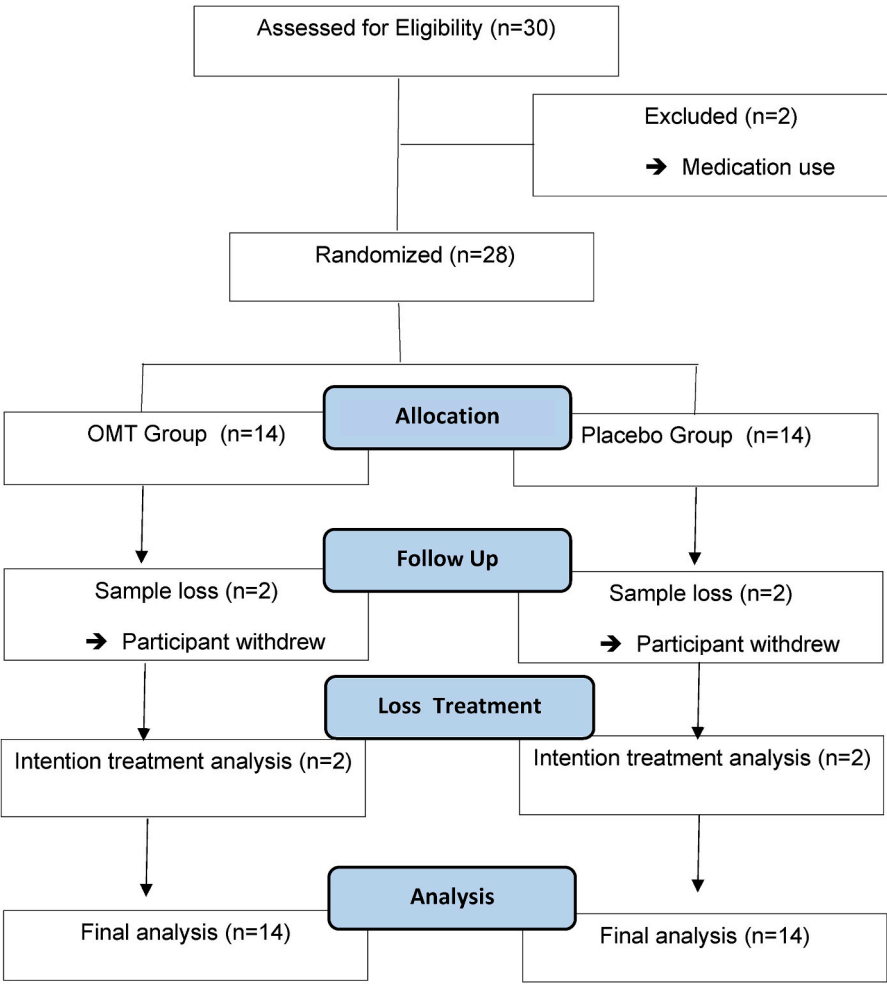


Fig. 1. Flow diagram of study.

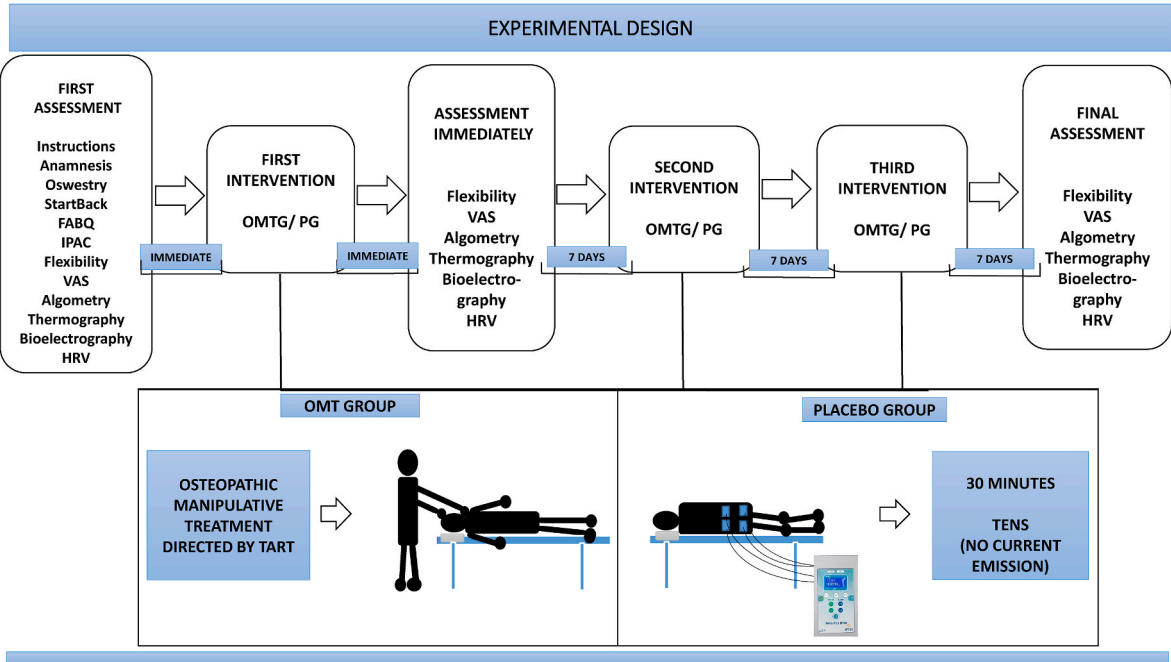


Fig. 2. Experimental design..

extension that has a ruler graduated in centimeters. For the measurement, volunteers remained seated, with their feet together and supported on the bench, and they were asked to slowly flex their trunk to the maximum, keeping their legs extended, until reaching maximum amplitude with their hands overlapping. The evaluator observed which point on the ruler the volunteer reached with their middle fingers in centimeters, as already referenced in previous studies by Ribeiro and collaborators (2018). Three measurements were taken and the average was used for analysis.

Heart rate variability was captured for 10 min, in the supine position, using the Polar RS800cx® heart rate monitor. Collection belt was properly attached to the volunteer's chest, the collection room was artificially air-conditioned, with the temperature maintained at 22 and 24 °C, and the experiment was carried out at the same time of day (2:00 p.m. to 6:00 p.m.) to minimize influences of circadian variations. Furthermore, volunteers were instructed to wear comfortable clothes on the day of the test, not to drink alcoholic beverages and/or stimulants (coffee, tea, energy drinks, etc.), avoid heavy meals and have a light meal at least 2 h before the test, and not to perform strenuous exercises 48 h before the assessments (Silva et al., 2017).

Analysis of heart rate variability was performed in the time domain (from heart rate data and intervals captured beat-to-beat) and frequency (spectral analysis) (Vanderlei et al., 2009). RR interval signals were transferred on Kubios HRV Standart HRV analysis program and a stable, artifact-free series of 256 RR interval sequences were selected and analyzed. Segments that showed any abnormalities were discarded. The analysis section selected for calculating the variables was 8 min, out of a total of 10 min of data collection. The first and last ones from the collection were excluded. In time domain, square root of the mean square index of differences between adjacent normal RR intervals (RMSSD) was used, which represents parasympathetic activity in heart rate control. In frequency domain, spectral analysis was performed, which provided for analysis the decomposition of periodic heart rate oscillations into pre-established frequency bands. The bands evaluated were low and high frequency, in normalized units, and the low/high frequency ratio (Vanderlei et al., 2009).

Thermographic evaluation was carried out by infrared images collected by a thermal imager (FLIR® Model OnePro; FLIR Systems, Inc., Oregon, USA). This device has an integrated resolution of 160 x 120 pixels, making it possible to measure temperatures between –20 °C and 400 °C, with thermal sensitivity of <0.10 °C.

To assessment, thermographic camera was positioned in the previously air-conditioned room, where the volunteer remained for 10 min at 22 °C, so that acclimatization could occur (De Souza Pinto et al. 2018). Everyone remained standing in front of the evaluator, with the area to be evaluated naked, with their feet on a rug at a distance of 2 m from the thermal imager. An image of the dorsal region was captured (Polidori et al., 2018). Image analysis was performed using the flir tools software, and the selection of interest area was considered the lumbar region, following a protocol previously described in the literature, and using average temperature for analysis. The region of interest (ROI) was selected by marking the level of the first lumbar vertebra (L1) to the level of the last lumbar vertebra (L5), considering for analysis the total area of the lumbar zone, right and left sides respectively (Ludwing et al. 2014).

Equipment used to obtain bioelectrographic photos was the *Bio-Well*, produced by the bio-well company. It has an electric current with an impulse duration of 10 µs, and repetition frequency of 1000Hz, with induction interval of 0.5–32 s, an electrode voltage of 3–15 kV, emitting a pulse of weak electric current for a very short period of time. Harmless to health, this current generates an electromagnetic field inducing the emission of photons and electrons from the skin, forming a luminous effect around the finger that is in contact with the equipment, with the luminous halo generated by this effect, called corona 70, which is the final product to be analyzed in bioelectrography (Korotkov, 2014). The photos obtained by Bio-Well were analyzed using *Bio-Well software*.

Total energy was measured in bioelectrography using *Bio-Well*, based on the corona captured by the camera, and is given by the formula (Korotkov, 2014): $E \text{ (joules)} = S \times I \times 4 \times 10^{-8}$, where E is the energy of the corona, S the area of the corona in pixels and I the energy in joules per cm² measured in the corona for the characteristics of the camera used in *Bio-Well*. The entropy coefficient measured by bioelectrography is the ratio between the external and internal contour of the corona. *Bio-Well* equipment is considered stable in terms of the data collected, since the variation in the electrographic photo parameters of a titanium cylinder, which is used in the *Bio-Well* calibration, is less than 1 % during the night and less than 3 % during the day (Korotkov, 2014). The tip of the distal phalanx of each finger inserted into the equipment touched the filter that was in contact with the lens that took the photo. The filter was used in order to minimize psychosomatic influences on the image obtained, seeking to evaluate the effect of osteopathic treatment as previously described (Korotkov et al., 2012).

3. Treatment protocol

The osteopathic manipulative treatment was carried out by an experienced osteopath, Diploma in Osteopathy, and member of the Brazilian Association of Osteopaths. The evaluation was guided by the changes found following the acronym TART (T - tenderness, A - asymmetry, R - restriction, T - texture changes) increased sensitivity, asymmetry, restriction and changes in tissue texture, thus identifying the somatic dysfunctions which led to the osteopathic intervention performed. Somatic dysfunctions were detected throughout the body, and then the osteopath defined the order of treatment according to the TART parameters. For each participant, the osteopath used an osteopathic note form (subjective, objective, assessment, plan). OMT techniques were centered on correcting dysfunctions found during the initial osteopathic assessment (at each session) and included myofascial techniques, structural, visceral and cranial techniques (Tramontano et al., 2020). Primary dysfunctions were selected by inhibition tests between dysfunctions.

For the placebo group, volunteers received an intervention without proven therapeutic effects. For this, the volunteers were positioned in lateral decubitus, with their thighs and legs flexed with a support between the legs to better accommodate the lumbar region. Four electrodes (2 on the right side and two on the left side) were fixed with adhesive tape to the lumbar paravertebral region of the volunteers. These electrodes were connected to an electrotherapy device (transcutaneous electrical nerve stimulation – TENS), in which the device was turned on but did not emit electrical current during the intervention (Khadiolkar et al., 2008; Contreras, 2013; Johnson et al., 2022).

3.1. Statistical analysis

All results were described as means and their respective standard deviations, and for all analyses, a significance level of 5 % was adopted. All statistical procedures were performed using the SPSS version 20.0 application.

To carry out the intention to treat analysis, missing data was imputed using the multiple imputation method (Armijo-Olivo et al., 2009), where 5 imputations were created for each variable, and for each volunteer, using the average of these values obtained for subsequent analysis.

After analyzing the distribution of data, the Shapiro-Wilk test was performed, with the data considered normal and the hypothesis of homogeneity of variances was evaluated using the Levene Test, thus, parametric tests were used to determine the significance of the data. For frequency comparisons, the Chi square Test was used. One-way ANOVA was performed for intergroup comparison for sample characterization data. Considering the hypothesis of interest of the study, to verify the intergroup differences throughout the pre, intermediate and post-training moments of the other variables, an analysis of variance

(ANOVA) was carried out in a mixed design (Group X Time). Multiple comparisons were performed with Bonferoni adjustment.

To analyze the relationship between variables, the Pearson Correlation Coefficient was used, with degrees of insignificant correlation being adopted when it is between 0 and 0.3, low between 0.3 and 0.5, moderate between 0.5 and 0.7, high between 0.7 and 0.9, and very high between 0.9 and 1 (Mukaka, 2012).

To calculate the effect size, the Cohen's *d* (Effect Size) method was used, calculated using the "Effect Size Generator" application, version 2.3 (Swinburne University of Technology, Center for Neuropsychology, Melbourne, Australia). Values greater than or equal to 0.8 represented a large effect size, between 0.2 and 0.8, a moderate effect and less than 0.2, a small effect (Cohen, 1988). To calculate the absolute and percentages data of body location from somatic dysfunctions evaluated in the group that received osteopathic manipulative treatment, percentage of location from somatic dysfunctions presented represents the total number of each group of dysfunctions (upper cervical, lower cervical for example) compared to the total number of somatic dysfunctions found in the evaluation OMT group.

4. Results

Characteristics of sample regarding age, anthropometric variables, scores from Oswestry, FABQ, StartBack questionnaires, as well as the level of physical activity indicated in IPAC are presented in Table 1. OMTG was significantly older than the PG. No significant differences were observed between the two groups for the other variables analyzed.

Table 2 presents the absolute and percentages data of body location from somatic dysfunctions evaluated in the group that received osteopathic manipulative treatment.

Table 3 presents characteristics relating to low back pain evaluated for both groups, as well as a comparison between the osteopathic manipulative treatment group and the placebo group.

Results for pain, flexibility and temperature are presented in Table 4. Significant Group $\times$ Time interaction was observed for flexibility variables ($F = 7.5$; $p = 0.001$) and VAS ($F = 2.8$; $p = 0.05$). Multiple comparisons revealed that there was a significant increase in flexibility between the pre and intermediate moments, as well as pre and post intervention only for the OMTG. As for VAS, there was a significant decrease in the pain ladder between the pre and intermediate moments for both groups, as well as a significant decrease in the VAS and VAS of the last week between the pre and post treatment moments for the OMTG, with a significant difference in the comparison between the groups for the post-treatment moment and moderate and high effect sizes, respectively.

For pressure pain, the lumbar algometry at L3 level showed a significant increase in the pressure threshold between the intermediate and post-treatment moments only in OMTG, with a moderate effect size observed.

Results of heart rate variability and bioelectrography are presented

Table 1
Sample characterization at beginning of experimental procedure.

Variables	OMTG	PG	P value
Age (years)	40.5 $\pm$ 3.5	35.7 $\pm$ 7.5	0.03
Weight (kg)	85.2 $\pm$ 17.1	83.5 $\pm$ 19.7	0.37
Height (m)	1.72 $\pm$ 0.7	1.70 $\pm$ 0.6	0.52
BMI (kg/m ²)	28.2 $\pm$ 4.3	28.0 $\pm$ 5.6	0.43
Score StartBack	3 $\pm$ 1.8	3.5 $\pm$ 1.5	0.51
Score FABQ	20.5 $\pm$ 11.6	23 $\pm$ 13.9	0.81
Score Oswestry	7.5 $\pm$ 4.7	9.5 $\pm$ 3.8	0.17
IPAC Active	9 (65 %)	6 (43 %)	0.43
IPAC Irregular Active A	1 (7 %)	1 (7 %)	>0.99
IPAC Sedentary	4 (28 %)	7 (50 %)	0.36

OMTG= Osteopathic manipulative treatment group; PG=Placebo group; BMI = body mass index; IPAC = international physical activity questionnaire.

Table 2

Location of somatic dysfunctions found in OMTG and primary dysfunctions.

Location of somatic dysfunctions	N (%)	Primary dysfunctions	N (%)
Skull	6 (6.1)	Skull	2 (14.2)
Upper cervical	10 (10.3)	Neck viscera	1 (7.1)
Lower cervical	5 (5.1)	Thoracic viscera	3 (21.4)
High chest	11 (11.3)	Abdominal viscera	2 (14.2)
Middle thoracic	7 (7.2)	Pelvic viscera	1 (7.1)
Low thoracic	9 (9.2)	Pelvis	1 (7.1)
High lumbar	2 (2.1)	Lower members	4 (28.5)
Low lumbar	8 (8.2)	Total	14 (100)
Pelvis	12 (12.3)		
Pelvic viscera	3 (3.1)		
Thoracic viscera	5 (5.1)		
Abdominal viscera	8 (8.2)		
Lower members	8 (8.2)		
Diaphragm	1 (1.1)		
Total	95 (100)		

OMTG= Osteopathic manipulative treatment group; N = absolute number; % = percentage.

Table 3

Characterization and comparison of low back pain for both groups.

Lumbar symptom time	N (%) OMTG	N (%) PG	P value
Up to 1 year	2 (14.2)	2 (14.2)	>0.99
1–3 years	5 (35.7)	4 (28.5)	0.73
3–5 years	3 (21.4)	2 (14.2)	0.65
>5 years	4 (28.5)	6 (42.8)	0.52
Lumbar symptom local	N (%) OMTG	N (%) PG	P value
Lower Right Lumbar	2 (28.5)	2 (28.5)	>0.99
Lower Left Lumbar	3 (21.4)	3 (21.4)	>0.99
Bilateral Lumbar	6 (42.8)	9 (64.2)	0.43
Irradiation for LE	1 (7.1)	0 (0)	>0.99

OMTG= Osteopathic manipulative treatment group; PG=Placebo group; N = absolute number; % = percentage; LE = lower extremity.

in Table 5. Significant Group $\times$ Time interaction was observed for the variables of LF ($F = 3.5$; $p = 0.04$), LF/HF ($F = 3.6$; $p = 0.04$) and total energy ($F = 4.4$; $p = 0.01$). Multiple comparisons revealed that there was a significant increase in RMSSD and TE between pre and post moments for OMTG with a moderate effect size for both. For immediate post-moment, an increase in RMSSD was observed for OMTG, and a reduction in LF and LF/HF in comparison between groups for OMTG. LF and LF/HF showed a significant reduction when comparing the pre and post intervention moments for OMTG, with difference between the groups at the post intervention moment and a high effect size. HF showed a difference between the groups, with an increase post intervention, and a high effect size. Total energy showed a significant increase in the intermediate and post intervention moments for OMTG with a moderate effect size.

Table 6 presents data from correlation analysis between variables of pain and flexibility in correlation with variables of total energy and heart rate variability. For correlation analysis, the absolute value of variation between the pre and post intervention moments was used.

Low positive significant correlation was observed between VAS versus LF and LF/HF for OMTG, indicating that negative VAS magnitudes were associated with negative LF and LF/HF magnitudes. For the PG, the 1-week VAS showed a low significant inverse correlation with the LF, indicating that negative magnitudes of the LF were associated with the positive magnitudes of the 1-week VAS, as well as the 1-week VAS showed a moderate significant correlation with the HF, indicating that positive magnitudes results from the 1-week VAS were associated with positive HF magnitudes.

Fig. 3 presents correlation between flexibility versus VAS and flexibility versus L3 algometry for PG. Fig. 4 presents same correlation for OMTG. A moderate positive significant correlation was observed

Table 4

Comparison of pain, flexibility and temperature between different moments (before, immediately after and after treatment).

Variables	Pre OMTG	PG	Post Immediate OMTG	PG	Post OMTG	PG	Effect GxT	ES
Flexibility (cm)	24.1 ± 9.6	26.9 ± 7.9	26.6 ± 7.9 *	27.6 ± 7.6	28.4 ± 8.3 [#]	26.7 ± 7.6	p = 0.001/F = 7.5	0.21
VAS (0–10)	4.5 ± 1.8	4.3 ± 1.6	1.8 ± 1.5 *	2.7 ± 1.5 *	1.7 ± 2.3 ^{#T}	3.8 ± 3.1	p = 0.05/F = 2.8	0.76
VAS 1 wk (0–10)	6.1 ± 2.1	6.3 ± 1.7	–	–	2.9 ± 2.8 ^{#T}	4.9 ± 2.1	p = 0.06/F = 2.9	0.80
Algometry L1 (kgf)	5.1 ± 1.7	5.3 ± 1.3	4.9 ± 1.1	5.1 ± 1.3	5.1 ± 1.2	5.6 ± 1.4	p = 0.70/F = 0.28	0.38
Algometry L2 (kgf)	5.1 ± 1.8	5.2 ± 1.5	5.3 ± 1.2	5.3 ± 1.4	5.5 ± 0.9	5.4 ± 1.4	p = 0.88/F = 0.12	0.10
Algometry L3 (kgf)	4.9 ± 1.8	5.3 ± 1.3	5.0 ± 1.2	5.1 ± 1.4	5.8 ± 1.1 **	5.4 ± 1.1	p = 0.24/F = 1.4	0.37
Algometry L4 (kgf)	4.5 ± 1.7	5.1 ± 1.3	4.8 ± 1.6	5.2 ± 1.2	4.8 ± 1.7	5.4 ± 0.9	p = 0.96/F = 0.3	0.45
Algometry L5 (kgf)	4.5 ± 1.5	5.2 ± 1.3	4.8 ± 1.3	4.8 ± 0.9	5.1 ± 1.2	5.2 ± 1.2	p = 0.26/F = 1.3	0.10
Thermography (°C)	31.9 ± 1.4	31.9 ± 1.6	31.8 ± 1.3	31.4 ± 2.2	31.4 ± 0.8	31.8 ± 1.5	p = 0.62/F = 0.6	0.34

OMTG= Osteopathic manipulative treatment group; PG=Placebo group; L = Lumbar zone; VAS = visual analogue scale; kgf = kilogram-force; *p < 0.05: comparison between pre and intermediate moments; **p < 0.05: comparison between intermediate and post moment; #p < 0.05: comparison between pre and post treatment; T p < 0.05: comparison between groups; GxT effect = time group effect; ES = effect size.

Table 5

Comparison of heart rate variability and bioelectrography variables, between moments (before, immediately after and after treatment).

Variables	Pre OMTG	PG	Post Immediate OMTG	PG	Post OMTG	PG	Effect GxT	ES
RMSSD (ms)	37.1 ± 18.8	32.4 ± 14.7	50.7 ± 20.2 *	38.7 ± 21.3	51.8 ± 22.5 [#]	41.8 ± 15.4	p = 0.62/F = 0.47	0.52
HF (u.n.)	49.5 ± 18.3	45.5 ± 11.9	55.7 ± 13.1	41.3 ± 9.8	60.8 ± 8.8 ^T	43.7 ± 12.3	p = 0.08/F = 2.6	1.59
LF (u.n.)	50.4 ± 18.2	53.4 ± 14.1	44.4 ± 13.3	58.6 ± 10.1 ^T	38.7 ± 8.7 ^{#T}	56.6 ± 12.7	p = 0.04/F = 3.5	1.64
LF/HF (u.n.)	1.3 ± 1.1	1.3 ± 0.7	0.9 ± 0.5	1.4 ± 0.5 ^T	0.7 ± 0.3 ^{#T}	1.4 ± 0.6	p = 0.04/F = 3.6	1.47
T Energy (J)	53.1 ± 7.1	58.3 ± 5.7	58.2 ± 8.1 *	59.5 ± 5.9	58.0 ± 3.9 [#]	54.8 ± 8.7	p = 0.01/F = 4.4	0.49

OMTG= Osteopathic manipulative treatment group; PG=Placebo group; RMSSD = root mean square of successive differ; HF = high frequency; LF = Low frequency; LF/HF = low/high frequency ratio; TE = total energy; ms = millisecond; u.n. = normalized unit; J = Joules; *p < 0.05: comparison between pre and intermediate moments; **p < 0.05: comparison between intermediate and post moment; #p < 0.05: comparison between pre and post treatment; T p < 0.05: comparison between groups; GxT effect = time group effect; ES = effect size.

Table 6

Correlations of absolute variations between pre and post intervention moments for both groups. Variables of pain and total energy were correlated, and pain with variables of autonomic heart rate modulation.

Correlations	Total Energy (J)	RMSSD (u.n.)	LF (u.n.)	HF (u.n.)	LF/HF (u.n.)
OMTG					
VAS (0–10)	r = −0.85 (p = 0.38)	r = −0.07 (p = 0.39)	r = 0.46 (p = 0.04)*	r = −0.44 (p = 0.05)*	r = 0.40 (p = 0.07)
VAS 1 wk (0–10)	r = −0.05 (p = 0.43)	r = 0.40 (p = 0.07)	r = 0.13 (p = 0.48)	r = 0.03 (p = 0.45)	r = −0.08 (p = 0.39)
PG					
VAS (0–10)	r = −0.23 (p = 0.21)	r = −0.09 (p = 0.37)	r = −0.38 (p = 0.09)	r = −0.33 (p = 0.13)	r = −0.34 (p = 0.11)
VAS 1 wk (0–10)	r = −0.05 (p = 0.43)	r = −0.38 (p = 0.09)	r = −0.50 (p = 0.03)*	r = 0.53 (p = 0.02)*	r = −0.43 (p = 0.06)

between flexibility and L3 algometry for OMTG, indicating an association between greater magnitude of flexibility variation with greater magnitude of pressure variation in L3.

Fig. 5 shows the correlation of absolute variations between the pre and post intervention moments, between variables of Total Energy versus the LF/HF ratio for both groups, indicating a low inverse correlation between the variables for OMTG.

5. Discussion

The main results pointed to significant impacts of OMT on increased flexibility, increased parasympathetic activity in the autonomic nervous system (ANS), increased total energy and reduced pain in volunteers with chronic low back pain. Furthermore, a relationship was observed between the magnitude of pain reduction and increased flexibility, as well as ANS modulation. After searching the literature, no study was found that investigated the effects of OMT in patients with chronic low back pain on associated mechanical, neurological and biofield electrophotonic imaging variables.

Improvement in flexibility was observed significantly immediately after the first intervention (24.1–26.9 cm) and also one week after the last intervention (24.1–28.4 cm), with a moderate effect size (0.21) one week after the third intervention for OMTG. These results were also

demonstrated in a systematic review of the literature, which indicated that osteopathic treatment promoted an increase in trunk flexion mobility parameters and functionality in patients with chronic low back pain (Franke et al., 2014) and also in a recent study that investigated the effect of OMT on posterior chain flexibility after 6 intervention sessions over 30 days (Zago et al., 2020).

The increase in parasympathetic activity in the ANS that was found in this study, demonstrated by the significant increase in RMSSD (37.1–51.8 ms; ES 0.52) and LF/HF (1.3 to 0.7 u.n.) for OMTG after the three treatment sessions, agrees with Ruffini et al. (2015), who intervened with osteopathic treatment in healthy volunteers observing effects on the ANS, and with Borges et al. (2018), who demonstrated in a systematic review the effects of manual treatment on the modulation of the ANS. With emphasis on chronic low back pain, these effects were also observed by Cerritelli et al. (2020), and by Cerritelli et al. (2021), who applied OMT to volunteers with chronic low back pain and found correlations between HRV modulations and the perfusion of brain areas related to pain, demonstrating that OMT with a global approach contributes to the adjustment of the ANS and modulation of pain in patients with chronic low back pain.

The increase in total energy (J) evaluated by biofield electrophotonic imaging observed in our results after OMT for OMTG (53.1–58.0 J; ES 0.49) was also found by other authors, such as Hossu and Rupert, 2006

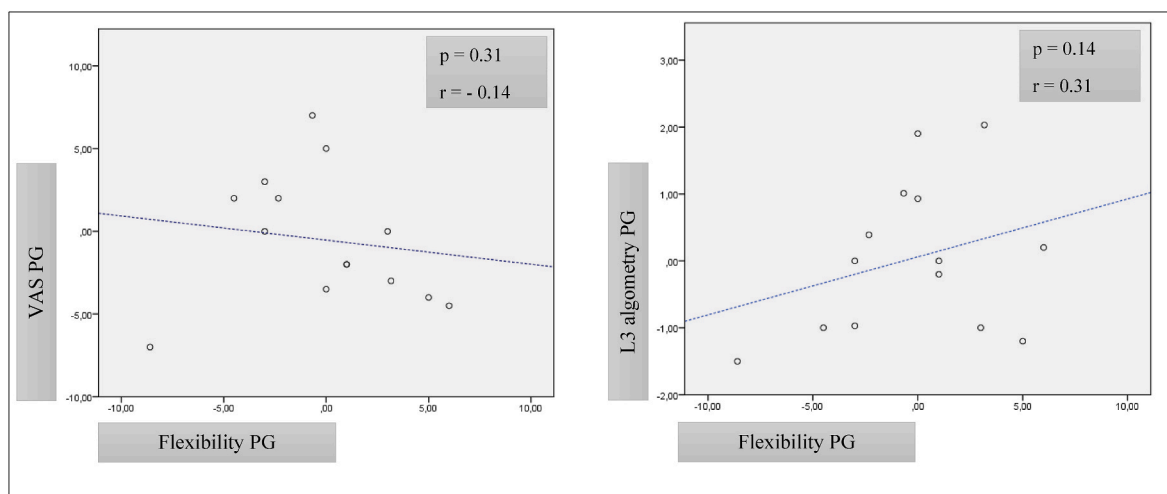


Fig. 3. Correlation of absolute variations between pre and post intervention moments, between variables of Flexibility versus VAS and Flexibility versus Algometry at the L3 level for the PG.

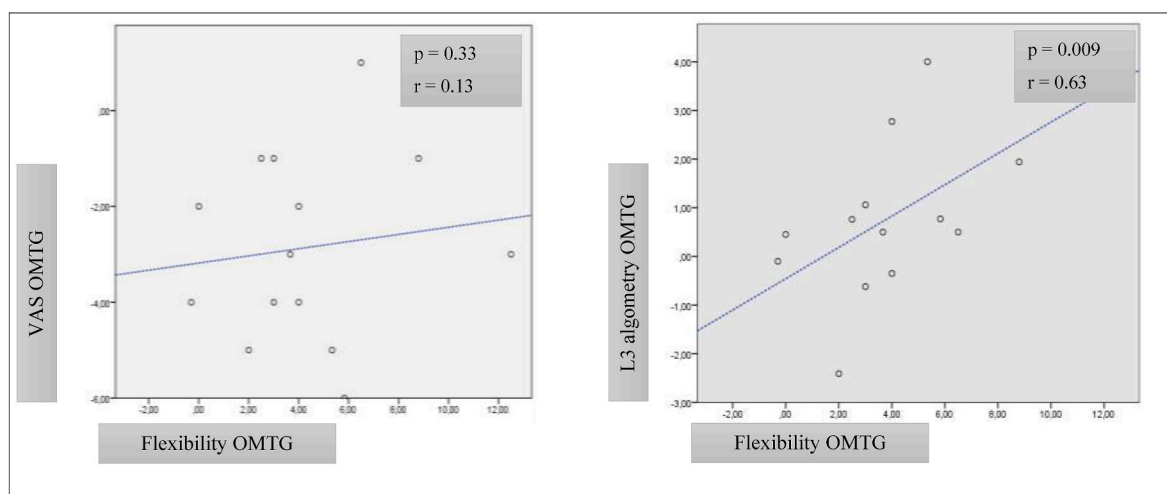


Fig. 4. Correlation of absolute variations between pre and post intervention moments, between variables of Flexibility versus VAS and Flexibility versus Algometry at the L3 level for the OMTG.

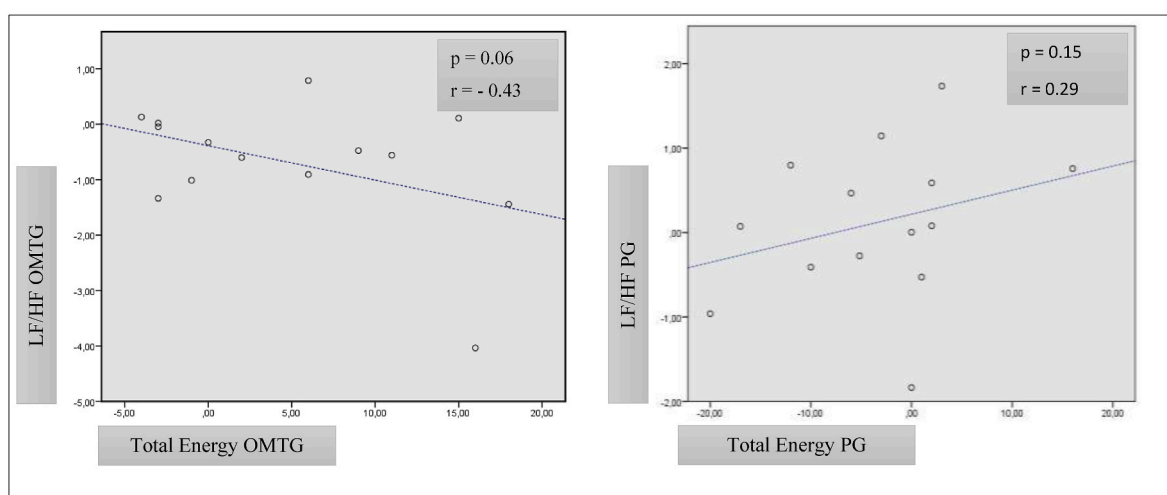


Fig. 5. Correlation of absolute variations between the pre and post intervention moments, between variables of Total Energy versus LF/HF ratio for both groups.

who found changes in the emission of biophotons after joint manipulation and craniosacral therapy and [Korotkov et al. \(2012\)](#), who found an improvement in the biofield electrophotonic imaging of healthy volunteers after OMT. Notably, this significant increase in biofield after OMT in the OMTG group in our study leads to the understanding that neural modulation also had an impact on biofield, an association observed in our results which pointed to a low inverse correlation between the total energy variable and the low frequency/high frequency ratio of autonomic heart rate modulation. These results corroborate the literature, which describes those increased changes in the biofield by electrophotonic imaging point to beneficial changes in autonomic modulation and may represent a marker of health and the individual's emotional state ([Korotkov et al., 2012](#)).

The reduction in pain observed corroborates [Licciardone et al. \(2016\)](#), who found reduced pain and improved functionality in volunteers with chronic low back pain after 6 OMT sessions, as well as [Franke et al. \(2014\)](#), who published a systematic review that showed positive effects in reducing pain and functional improvement in volunteers with acute and chronic nonspecific low back pain. [Dal Farra et al. \(2021\)](#), who also published a systematic review that showed improvement in pain and function after OMT that used fascial, manipulation, craniosacral and visceral treatment techniques.

The biopsychosocial model of chronic pain says that pain is the consequence of the complex and dynamic interaction of biological, psychological and social aspects. Therefore, the tissues that make up the lumbar spine are prone to different stressors, and these, alone or in combination, can contribute to low back pain. ([Knezevic et al., 2021a,b](#)).

In the context of nonspecific chronic low back pain, nociplastic pain stands out, which is defined as pain resulting from the altered function of sensory pathways related to pain in the periphery and in the central nervous system, causing increased sensitivity, which can occur in isolation or associated with chronic pain conditions that are primarily nociceptive or neuropathic ([Fitzcharles et al., 2021](#)). They share common neurophysiological mechanisms involving several systems and organs ([Nicholas et al., 2019](#)).

Therefore, for the diagnosis and treatment of individuals with nociplastic pain, there is a need for a comprehensive assessment, considering pain characteristics, fatigue, cognitive problems, multiple environmental problems, sensitivities and psychological signs, as well as a broad physical examination, covering various tissues and systems ([Fitzcharles et al., 2021](#)).

In this context, somatic dysfunction is defined as an impaired or altered function of related components of the somatic system, including related skeletal, arterial, myofascial, vascular, lymphatic and neural structures ([Weston et al., 2011](#)). The osteopathic manipulative treatment approach aims to find somatic dysfunction through a global osteopathic palpatory diagnosis, using different manual techniques to restore somatic dysfunction, which is achieved by restoring normal tissue movement and consequent body functionality ([Seffinger et al., 2018](#)). As a result, we observed that the volunteers assessed in our study had different primary dysfunctions, involving different regions of the body: lower limbs (28.5 %), thoracic viscera (21.4 %), abdominal viscera (14.2 %), skull (14.2 %), neck viscera (7.1 %), pelvis (7.1 %), respectively. This information shows a variance in the predominance of somatic dysfunctions, involving mechanical alterations in the lower limbs and pelvis, followed by visceral dysfunctions in the volunteers assessed, which helps us to understand the need to carry out an individualized assessment and intervention treatment.

It is worth noting that the global assessment and assessment of different systems is the basis for the osteopathic diagnosis and this is justified by the different treatment sites that each volunteer received. A correlation was found between decreased pain and increased flexibility of the posterior muscular chain and also between pain and increased parasympathetic activity, which indicates that OMT generated biomechanical and neurological responses in volunteers, positively influencing the balance of the ANS and, consequently in interoception,

which, when dysfunctional, contributes to potentiating nociplastic pain.

The increase in flexibility was probably due to the influence of the global treatment on the thoracolumbar fascia, which has connections that interconnect the trunk, upper and lower limbs and plays an important role in the transmission of tension, functional balance and stabilization of the lumbar spine. It is worth mentioning that individuals with chronic low back pain present loss of function and greater predominance of the thickness of the deep layer of the thoracolumbar fascia ([Marpalli et al., 2021](#)), as well as reduced sliding between layers of the thoracolumbar fascia and increased nociceptive function. ([Tesarz et al., 2011](#), [Suarez-Rodriguez et al., 2022](#)). In this sense, approaches that seek to reduce the density and stiffness of fascial tissue can increase flexibility and, consequently, reduce pain.

The increase in parasympathetic activity positively influenced the improvement of pain. Patients with chronic low back pain present sympathetic predominance and a decrease in the parasympathetic component ([Bandeira et al., 2021](#)). HRV can be considered an indirect marker of the physiological measure of homeostasis capacity and neurogenic regulation ([Bandeira et al., 2021](#); [Cerritelli et al., 2021](#)). Considering that the neural system configures a complex organization that integrates signals from inside and outside the body and adaptively regulates cognitive and behavioral responses, perception and action, including sensory and motor changes (Trayer et al., 2000), its regulation contributes to the modulation of these responses.

Especially in the chronic pain scenario, the evaluation and measurement of variables becomes challenging due to the complexity of interactions that influence pain complaints. However, recent studies have observed correlations between HRV modulations and perfusion of pain-related cortical areas, assessed by functional magnetic resonance imaging [Cerritelli et al. \(2020, 2021\)](#). These studies demonstrate the significant interaction of the brain-heart mechanism, observing correlations between measures of autonomic modulation and brain activity in volunteers with chronic pain, as well as observing effects of OMT on the reduction of activity in areas related to interoception, especially the insular cortex and anterior cingulate cortex, associated with improved autonomic modulation of heart rate, demonstrating its potential to modulate brain activity in chronic patients.

Based on the information above we believe that the improvement in pain induced by OMT in our results, which showed a significant correlation with the parasympathetic activity of the OMT group, can be justified by the broad treatment approach, which addressed different systems and was focused on the individual needs of each volunteer, thus contributing to the increase in homeostatic capacity, achieved by modulating the neurological system. Furthermore, our results demonstrated significant correlation between flexibility and L3 algometry for OMTG, so, improvement in pain probably occurred because of improved mobility, which reduces the overload of mechanical component on the lumbar spine and the modulation of the autonomic nervous system, reducing the viscerosomatic reflex ([Ceballos-Laita et al., 2023](#)), contributing to regulating the perception of pain.

OMT uses craniosacral, visceral and structural intervention, applied alone or in combination ([Seffinger et al., 2018](#)). The effectiveness of craniosacral treatment on pain and functional disability in patients with short and long term chronic pain, including chronic low back pain, has also been demonstrated ([Haller et al., 2019](#)) and may be due to the normalization of autonomic activity achieved by modification of the craniosacral rhythm ([Cutler et al., 2005](#); [Fornari et al., 2017](#)).

Likewise, visceral manual treatment contributes to the reduction of muscle spasm and fascial limitations in the lumbar spine by regulating the viscerosomatic reflex arc, as well as its effect on the emotional improvement of patients ([Ceballos-Laita et al., 2023](#)). [Tamer et al. \(2017\)](#) found improved quality of life, decreased physical limitations and increased energy in volunteers with chronic low back pain after 10 sessions of visceral treatment indicated according to each volunteer's needs. In our present study, around 49 % of the patients' primary dysfunctions were related to the visceral system, therefore, we believe that

the visceral treatment approach in our sample contributed to the reduction of the viscerosomatic reflex and consequent improvement in the patients' mobility and symptoms.

Based on the above, as a treatment approach for patients with chronic non-specific low back pain, which has a multifactorial characteristic and several impacted systems, OMT had a positive impact on mechanical (flexibility), neurological (pain, ANS) and associated bio-field energetic parameters. This contribution can help to understand the mechanisms of action of OMT, which favor the improvement of patients with chronic low back pain.

Considering the functionality component, previous studies (Licciardone et al., 2016; Dal Farra et al., 2021) investigated the impact of OMT on this outcome, and pointed to improvement in the group that received OMT in pain reduction and correlation with improved functionality. We observed in our results that the volunteers showed a significant reduction in pain, however the values found in the *Oswestry* questionnaire scores were used to characterize the sample showing low disability scores (score 7.5 for OMTG and score 9.5 for PG), thus limiting the interpretation of the effects of our intervention on the aspect of functionality, as well as demonstrating that even patients with low levels of disability can have chronic pain.

Finally, although we did not observe a significant interaction for lumbar temperature in our results, we observed a moderate effect size at the end of treatment indicating a reduction (31.9–31.4 °C, ES 0.34) in lumbar temperature for the OMTG. Although infrared thermography is applied to contribute to the non-invasive identification of changes in skin temperature, which can be suggested as a result of functional and/or muscle changes related to movement, thermography is commonly used as a complementary method for functional assessments (Côrte et al., 2019). The effect of manual therapies on skin temperature has been described previously, with the decrease in thermographic temperature and decrease in pain observed in patients with coccygodynia, and who received 12 weeks of local manipulative treatment, and high significant correlation was pointed out between the reduction in temperature and pain in the volunteers evaluated in prior study (WU et al., 2009). Polidori et al. (2018), found a decrease in temperature in the lumbar spine in volunteers with chronic low back pain after OMT with manipulative, myofascial and visceral techniques of the lumbar region, and credited these results to the reduction of tissue stress and the local inflammatory process. We believe that reducing temperature may have an influence on reducing pain and improving local tissue mobility, however, the interpretation of these findings requires further investigation.

As limitations of this study, we believe that comparison with an active treatment group could provide additional information to our results. Furthermore, although a positive response was found after one month of intervention, long-term follow-up could provide additional information about the effects of OMT. For future studies, it is suggested to replicate the assessments 3 months and 6 months after the interventions to obtain data after a longer period without intervention, as well as, although in this study we followed sample calculation, for new explorations we suggest increasing the number of volunteers for better interpretation of the observed results. In addition, the pragmatic treatment, despite showing “real world” results (Makaram and Simpson, 2023), makes it difficult to the standardization and replication of the study, which could interfere with the quantification and level of significance of the findings. Furthermore, the effects of OMT on volunteers with severe levels of disability and the changes in skin temperature obtained by thermography should be explored.

6. Conclusion

The osteopathic manipulative treatment consisting of three intervention sessions resulted in reduced pain, increased flexibility of the posterior chain, increased parasympathetic activity and increased total energy in patients with chronic low back pain. The reduction in pain,

increase in flexibility and parasympathetic activity as well as total energy were higher after three intervention sessions than after one session, which demonstrated the greater effectiveness of treatment with a greater number of interventions in this population.

CRedit authorship contribution statement

Charlini Simoni Hartz: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Marília Almeida de Molon Mendes:** Resources, Formal analysis, Data curation. **Kelmerson Henri Buck:** Methodology, Formal analysis, Data curation, Conceptualization. **Marlene Aparecida Moreno:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Gustavo Luiz Bortolazzo:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbmt.2025.06.003>.

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